

PATIENT FEATURES

OTP Email Verification	Register !' receive 6-digit OTP !' verify account
Secure Login / Logout	JWT access token + httpOnly refresh cookie
Forgot / Reset Password	Secure reset link sent to email (30 min expiry)
Profile Management	Name, avatar, blood group, chronic conditions
AI Health Chat	Conversational chat with Google Gemini AI
Voice Input	Speak queries — AI responds in text
Chat History	Multiple sessions, full message history
Medical AI Analysis	AI analyses uploaded health records
AI Pre-consultation Summary	AI summary generated before appointments
Browse Doctors	Search doctors by specialization
View Available Slots	See open time slots for any doctor on any date
Book Appointment	In-person / video / phone consultation
Cancel Appointment	Cancel with reason — doctor notified instantly
Appointment History	View all past and upcoming appointments
Upload Health Records	PDF, JPG, PNG documents
OCR Text Extraction	Extract text from scanned/image documents
PDF Parsing	Parse structured data from PDF reports
AI Health Analysis	Get AI insights from your health records
Notification Bell	Real-time bell with animated unread count badge
Notification Dropdown	Latest 10 notifications with type icons
Notifications Page	Full center with filters (All / Unread / by type)
Reminder Banner (1h)	Floating amber banner before upcoming appointments
Welcome Email	Sent on successful registration
OTP Email	Branded 6-digit code for verification
Password Reset Email	Secure link with 30-minute expiry
Appointment Confirmed Email	Sent when doctor confirms booking
Cancellation Email	Sent if doctor cancels your appointment
Self-cancel Confirmation	Sent when patient cancels their own appointment
24h Reminder Email	Sent automatically 24 hours before appointment

👤 Doctor Registration	Register with specialization, fees, availability
🕒 Availability Settings	Set start/end time and slot duration
💰 Consultation Fee	Set your consultation fee on profile
📅 View All Appointments	Dashboard with complete appointment list
✅ Confirm Appointment	Approve pending patient bookings
🏁 Complete Appointment	Mark appointment as completed
❌ Cancel Appointment	Cancel with reason — patient notified + email sent
📝 Add Doctor Notes	Attach clinical notes to any appointment
🔔 Notification Bell	Real-time bell with unread badge
📧 New Booking Alert	Instant notification when patient books
📧 Cancellation Alert	Notification when patient cancels
#📢 Reminder Banner (1h)	Floating banner for upcoming appointments
✉️ New Booking Email	Styled email with patient name, date, time, reason
✉️ Patient Cancellation Email	Email when a patient cancels their booking
✉️ 24h Reminder Email	Auto-sent 24 hours before each appointment
✉️ 1h Reminder Email	Auto-sent 1 hour before each appointment

⚙️ SYSTEM & AUTOMATION

- JWT access tokens (short-lived) + httpOnly refresh cookie for silent renewal
- Rate limiting: 300 req/15min global · 20 req/15min on auth routes
- NoSQL injection prevention (express-mongo-sanitize)
- HTTP parameter pollution protection (hpp)
- Security headers via Helmet
- CORS with whitelist of allowed frontend origins
- OTP codes auto-expire from DB after 10 minutes (MongoDB TTL)
- Notifications auto-expire from DB after 90 days (MongoDB TTL)
- node-cron: 24h reminder job runs every hour (0 * * * *)
- node-cron: 1h reminder job runs every 15 minutes (* / 15 * * * *)
- Duplicate reminder prevention via reminderSent24h / reminderSent1h flags
- Cron logs with [CRON] prefix in server console
- Google Gemini AI with model fallback chain for reliability
- Tesseract.js OCR — extracts text from images and scanned PDFs
- pdf-parse — structured data extraction from PDF documents
- Winston structured logging (JSON in production, colored in dev)
- Morgan HTTP request logging
- Centralized error handling middleware (AppError class)
- Input validation on all API routes (express-validator)
- Cloudinary integration for file/image storage

✉️ EMAIL TEMPLATES

Template	Recipient	Trigger
Welcome Email	Patient	On registration
OTP Verification	Patient	On OTP request
Password Reset	Patient	Forgot password
Appointment Booked	Doctor	Patient books an appointment
Appointment Confirmed	Patient	Doctor confirms
Appointment Cancelled	Patient / Doctor	Either party cancels
Self-cancel Confirmation	Patient	Patient cancels their booking
24h Reminder	Patient + Doctor	Auto — 24h before appointment
1h Reminder	Patient + Doctor	Auto — 1h before appointment

📄 PROJECT SUMMARY

